Apr. 16

- Thm. 12.1 Let $a, b$, and $c$ belong to a ring $R$.

1) $a0 = 0a = 0$

$0+0=0$ because $(R,+)$ is an Abelian group

If $a \in R$, then $a(0+0) = a0$, so $a0 + a0 = a0$ since mult. distributes over $+$

Subtracting $a0$ from both sides, $a0 = 0$. ✓

Note: we use $-r$ to denote the additive inverse of $r \in R$ and $s-r$ to denote $s+(-r)$

we will use the term subtraction

2) $a(-b) = (-a)b = -(ab)$

We want $a(-b) = -(ab) = -ab$

Strategy: show $a(-b)+ab=0$, and since the additive inverse is unique, we are done

Consider $ab + a(-b) = a(b+(-b)) = a(0) = 0$. Since additive inverses are unique, for groups

we see $a(-b) = -(ab)$. Similarly, $(-a)b = -(ab)$. ✓

3) $(-a)(-b) = ab$

4) $a(b-c) = ab - ac$ and $(b-c)a = ba - ca$

furthermore, if $R$ has unity element $1$, then

5) $(-1)a = -a$

We compute $a + (-1)a = 1 \cdot a + (-1) \cdot a = (1-1)a \overset{\star}{=} (0)a = 0$. Since additive inverses are unique, we conclude $(-1)a = -a$. ✓

★ Notice we factored out $a$ on the right, since we don't have general commutativity of multiplication, but $a(-1) = -a$ by a symmetric argument

6) $(-1)(-1) = 1$

- To prove: the mult. identity (unity), if it exists, is unique

Suppose $1$ and $e$ are both unities in a ring $R$. Then $1 = 1 \cdot 1 = 1 \cdot e = e$, so $1 = e$. Thus, the unity is unique.

- Def Let $R$ be a ring. A subring $S \subseteq R$ is a nonempty subset that is a ring under the same operations as $R$.

- Theorem 12.3 (The Subring Test) A nonempty subset $S \subseteq R$ is a subring if it is closed under subtraction and multiplication.

Proof We want: $(S,+)$ is an Abelian group, $(S,\cdot)$ is closed, associative, and distributive over $+$

By the 1-step subgroup test, $(S,+)$ is a subgroup of $(R,+)$ if it is nonempty and $ab^{-1} = a-b \in S$ whenever $a, b \in S$. Since $S \neq \emptyset$ and $S$ is closed under subtraction, the 1-step

Subgroup test shows $(S,+)$ is a subgroup of $(R,+)$. Since $a+b=b+a$ $\forall a,b\in R\supseteq S$, we see $(S,+)$ is Abelian. Let $a,b\in S$. Since $S\subseteq R$, we have $a,b\in R$. Since $(R,+)$ is Abelian, $a+b=b+a$. Thus $(S,+)$ is Abelian.

By hypothesis, $(S,\cdot)$ is closed under $\cdot$. Since $a(bc)=(ab)$ $\forall a,b,c\in R\supseteq S$, so $(S,\cdot)$ is associative. Let $a,b,c\in S$. Since $S\subseteq R$, we have $a,b,c\in R$. Since $(R,\cdot)$ is associative, $a(bc)=(ab)c$. Thus $(S,\cdot)$ is associative. Closure by hypothesis.

Let $a,b,c\in S$. Since $S\subseteq R$, we have $a,b,c\in R$. Since $R$ is a ring, $a(b+c)=ab+ac\in S$ since $S$ is closed under $\cdot$ and $+$. Likewise, $(b+c)a=ba+ca\in S$ since $S$ is closed under $\cdot$ and $+$. Thus $(S,\cdot)$ is distributive over $+$. $\square$

★No standard notation for subring, just say "is a subring of"

Ex. Consider $\mathbb{Z}[\frac{1}{2}]=\{\frac{a}{2^n} : a\in\mathbb{Z}, n\in\mathbb{Z}^{\geq 0}\}$ as a subset of $\mathbb{Q}$. Show it is a subring of $\mathbb{Q}$.

We will apply the subring test.

It is clearly nonempty.

Let $\frac{a}{2^n}$ and $\frac{b}{2^m}$ be arbitrary elements. We see $\frac{a}{2^n}\cdot\frac{b}{2^m}=\frac{ab}{2^{n+m}}\in\mathbb{Z}[\frac{1}{2}]$ since $ab\in\mathbb{Z}$ and $nm\in\mathbb{Z}^{\geq 0}$. (closure under $\cdot$)

Likewise, consider $\frac{a}{2^n}-\frac{b}{2^m}=\frac{a\cdot 2^m}{2^{n+m}}-\frac{b\cdot 2^n}{2^{n+m}}=\frac{a2^m-b2^n}{2^{n+m}}\in\mathbb{Z}[\frac{1}{2}]$ since $a2^m-b2^n\in\mathbb{Z}$ and $n+m\in\mathbb{Z}^{\geq 0}$. $\square$

- The notation $\mathbb{Z}[\frac{1}{2}]$ indicates the smallest★ ring containing $\mathbb{Z}$ and $\frac{1}{2}$
- The notation $\mathbb{Z}[x]$ indicates the smallest★ ring containing $\mathbb{Z}$ and $x$ (also polynomials in variable $x$)

★here, smallest refers to containment, not necessarily size/cardinality

## Ch.13 Integral Domains

"Rings that are even more like the integers"

not relating to $\int$, more like the integers

- Def A zero-divisor is a nonzero element $a$ of a ring $R$ such that $ab=0$ for some nonzero $b$.

- Def An integral domain is a commutative ring with unity (1) and no zero-divisors

Examples: $\mathbb{Z}, \mathbb{Z}[\frac{1}{2}], \mathbb{Z}[x], \mathbb{Q}, \mathbb{R}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}$ for $p$ prime

Non-examples:

- $\mathbb{Z}/6\mathbb{Z}$ since $\bar{2}\cdot\bar{3}=\bar{6}=\bar{0}$
- $M_2(\mathbb{R})$ since $\begin{bmatrix}1&0\\0&0\end{bmatrix}\cdot\begin{bmatrix}0&0\\1&0\end{bmatrix}=\begin{bmatrix}0&0\\0&0\end{bmatrix}$
- $\mathbb{Z}/n\mathbb{Z}$ for $n$ composite ($n=ab$) since $\bar{a}\cdot\bar{b}=\bar{n}=\bar{0}$
- $\mathbb{Z}\times\mathbb{Z}$: $(0,1)\cdot(1,0)=(0\cdot 1,1\cdot 0)=(0,0)$

- <u>Theorem 13.1</u> (Cancellation) Let $a, b, c$ be elements of an integral domain with $a \neq 0$. If $ab = ac$, then $b = c$.

  <u>Proof</u>

  If $ab = ac$, then $ab - ac = 0$. Hence $a(b-c) = 0$. Since $R$ is an integral domain, $a = 0$ or $b - c = 0$. Since $a \neq 0$ by hypothesis, $b - c = 0$. Thus $b = c$. $\square$